The Economic Impact of Village Funds and Pre-Existing BUMDes: Evidence from East Java

Freddy Sitorus
Columbia University in the City of New York

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Abstract

This paper investigates the role of institutional capacity in mediating the effectiveness of large-scale fiscal decentralization, using Indonesia's Village Fund policy as a case study. Leveraging a difference-in-differences design, I examine whether the presence of BUMDes prior to the 2015 policy implementation amplified the impact of village-level fiscal transfers on village income generation. Using nationally representative village-level data from the Village Potential Survey and administrative records from East Java province, I find that villages with pre-existing BUMDes experienced a 53% higher increase in log village income relative to those without, when provided with fiscal transfers from the central government, even after controlling for demographic and structural characteristics. Event study and parallel trend analyses confirm the validity of the identification strategy. Testing the heterogeneous effects reveals that this institutional advantage is most pronounced in medium-poverty villages and those led by more educated village heads, suggesting that both absorptive capacity and leadership quality are critical enabling factors. In contrast, the effects are weaker and more variable in high-poverty and large-population villages, suggesting the need for more tailored policy interventions; in the case of large populations, this may include scaling up funding to match greater resource demands. The implications extend beyond Indonesia, offering lessons for other countries confronting institutional fragility, where decentralization reforms have often fallen short due to overestimated assumptions about local implementation capacity. In such settings, coupling decentralization with targeted investments in institutional readiness is essential to achieving inclusive rural development.

Keywords: Village Fund Indonesia, Fiscal Decentralization, Institutional Capacity, BUMDes, Rural Development, Inequality, Village Income.

Motivation

Despite notable national progress, rural development continues to lag behind in Indonesia. As of 2024, the rural poverty rate remains at 11.34%, significantly higher than the urban poverty rate of 6.66%[1]. Inequality also persists, with a Gini coefficient still at approximately 0.4, highlighting the necessity for targeted policy interventions. Several targeted policy initiatives have been undertaken by the current and previous administrations, with the largest and most prominent being the introduction of the Village Fund policy in 2015. This policy represents a substantial scaling-up of direct fiscal transfers, reaching nearly all of Indonesia's more than 75,000 villages[2].

Nearly a decade later, the Village Fund has considerably contributed to village development, particularly noticeable in infrastructure improvements. However, economic outcomes vary significantly across villages. [3] Some villages have achieved substantial improvements, as reflected in increased village revenue and per capita income, while others have experienced minimal or very gradual progress since the policy's inception. This divergence suggests that although villages receive similar fiscal resources, additional factors are critical in effectively converting these funds into tangible economic outcomes and enhancing the welfare of rural populations, which is the ultimate objective of this policy.

This paper aims to identify and isolate one key enabler that allows villages to translate Village Fund resources into improved economic performance: institutional capacity, specifically the capability to facilitate and drive economic activities. I use the presence of Village-Owned Enterprises (Badan Usaha Milik Desa, or BUMDes) as a proxy for measuring institutional capacity. By comparing villages with existing BUMDes prior to the introduction of the Village Fund policy to those without, I analyze differences in the trajectories of village income generation. In the same way, I will also examine whether certain characteristics such as the initial poverty level before the Village Fund policy, the educational attainment of the village head, and the population size associated with different treatment effects on my primary outcome variable.

To assess this relationship, I apply a difference-in-differences (DiD) empirical strategy combined with an event study framework, leveraging panel data from the Village Potential Survey (Podes) covering four observation periods (2008, 2011, 2014 pre-treatment, and 2019 post-treatment). My findings demonstrate that villages with pre-existing BUMDes experienced approximately a 53% higher increase in log

[1]Badan Pusat Statistik. (2025, May 2). ***Understanding the differences in poverty rates reported by the World Bank and Statistics Indonesia (BPS)***. https://www.bps.go.id/en/news/2025/05/02/702/understanding-the-differences-in-poverty-rates-reported-by-the-world-bank-and-statistics-indonesia--bps

[2]Badan Pusat Statistik. (2024, December 10). ***Village potential statistics of Indonesia 2024***. https://www.bps.go.id/en/publication/2024/12/10/2f5217e2d6a695a0830290a7/village-potential-statistics-of-indonesia-2024.html

[3] Saputra, Rakhmat Nur Hakim. 2024. "Satu Dekade Dana Desa: Samar Jejak Ratusan Triliun Rupiah." ***Kompas.id***, May 9. https://www.kompas.id/artikel/satu-dekade-dana-desa-samar-jejak-ratusan-triliun-rupiah.

village income relative to those without, when provided with fiscal transfers from the central government. Additionally, heterogeneous effect analysis shows that these results are particularly robust in medium-poverty villages and those with better-educated village heads, suggesting both absorptive capacity and leadership quality as critical enabling factors.

The central argument is that financial transfers alone are insufficient; the government must also address the institutional capacity and readiness of villages to maximize the impact and cost-effectiveness of one of the nation's most substantial development policies (Grindle, 2007; Wunsch, 2001; Agrawal & Ribot, 1999).

Background

The Village Fund is mandated by Law No. 6 of 2014 on Villages, marking a major decentralization reform aimed at strengthening local governance. The policy is centered around four key objectives: reducing rural poverty and inequality, improving village infrastructure and basic services, enhancing village governance and administrative capacity, and promoting local economic development. This policy entails substantial fiscal transfers directly to local village authorities, providing them significant discretion in allocating funds according to their community-specific priorities and needs.

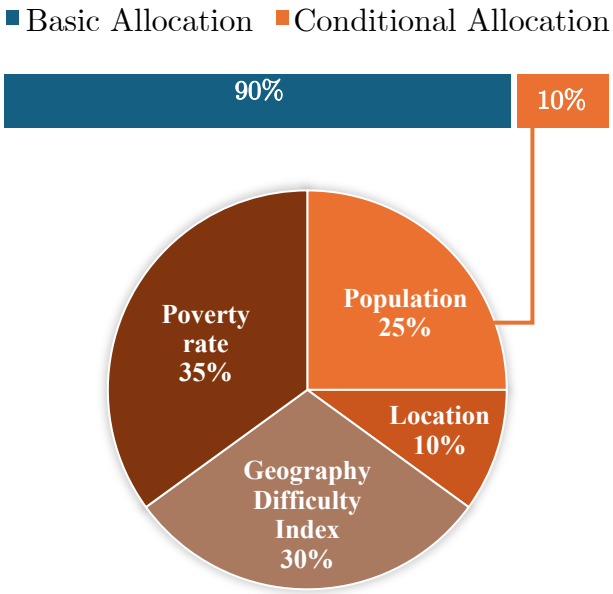
This decentralized funding mechanism closely resembles block grants commonly utilized in fiscal federalism, particularly in the United States. Block grants, such as the Community Development Block Grant (CDBG) and Temporary Assistance for Needy Families (TANF), offer financial resources from the federal government to states and local governments with broad guidance but considerable flexibility in allocation (Conlan, 1998; Dilger and Cecire, 2019). Like Indonesia's Village Fund, block grants aim to empower lower levels of government by enabling them to tailor spending to local circumstances, thus promoting responsiveness and effectiveness (Oates, 1999).

However, despite these similarities, some key distinctions are notable. While U.S. block grants typically operate at state or municipal levels and span diverse policy areas, Indonesia's Village Fund specifically targets rural villages with clear developmental objectives, including poverty reduction and economic stimulation through local institutions such as Village-Owned Enterprises (BUMDes). Moreover, the Village Fund incorporates specific reporting and accountability requirements to ensure effective use of funds, reflecting an approach adapted to Indonesia's unique institutional and governance context.

Understanding these parallels and differences contributes to broader discussions in fiscal federalism about how decentralization impacts development outcomes. It underscores the importance of local institutional capacity and governance quality, factors that this paper explores empirically by examining the effectiveness

of Village Funds through pre-existing BUMDes in East Java.

Figure 1. Village Fund Allocation
Source: Directorate General of Fiscal Balance



Historically, the fund has primarily been used for infrastructure investments such as roads, irrigation, clean water, and sanitation, as well as for supporting education, health, social protection programs, and cash transfers that stimulate economic activity at the village level.

As illustrated in **Figure 1**, the basic allocation of the fund is distributed evenly across villages, with variations based on demographic and geographic characteristics, including population size, poverty incidence, land area, remoteness, and performance metrics[4]. The fund is disbursed annually, and all recipient villages must submit an accountability and evaluation report to qualify for the following year's disbursement. The budget allocation has grown substantially from IDR 20.7 trillion in 2015 to over IDR 70 trillion annually throughout the 2020s, with over 74,000 villages receiving regular transfers[5]. The implementation has evolved over the decade, including adjustments during the COVID-19 pandemic to prioritize labor-intensive cash-for-work programs, and more recent efforts to introduce performance-based allocation mechanisms. The government also launched monitoring tools such as the Village Development Index (Indeks Desa Membangun) and Siskeudes (the village financial system) to improve oversight and accountability[6]. Despite positive developments in infrastructure and civic participation, the program continues to face challenges, including mismanagement, elite capture, and capacity constraints, particularly in more remote areas (Permatasari et al., 2024).

Badan Usaha Milik Desa (BUMDes), or village-owned enterprises, are community-based business entities owned and managed by village governments with the purpose of increasing residents' welfare through the facilitation of economic activity. While some BUMDes were established prior to

[4] World Bank. (2019). ***Village public expenditure management in Indonesia: Towards better budgeting and spending.*** https://documents1.worldbank.org/curated/en/689971628748237085/pdf/Village-Public-Expenditure-Management-in-Indonesia-Towards-Better-Budgeting-and-Spending.pdf

[5] Kompas. (2025, March 15). ***A decade of village funds, traces of hundreds of trillions of rupiah.*** https://www.kompas.id/artikel/en-satu-dekade-dana-desa-samar-jejak-ratusan-triliun-rupiah

[6] World Bank. (2022). ***Institutional strengthening for improved village service delivery project.*** https://documents1.worldbank.org/curated/en/099121524221036020/pdf/P1655431359d5c0c31bec2174836aa799dd.pdf

2014, the legal foundation for their broader institutionalization was provided through Law No. 6 of 2014, which formally empowered villages to manage their own resources and drive localized economic initiatives.

The main objectives of BUMDes include mobilizing local resources to stimulate economy and employment, generating additional revenue for villages, and improving residents' quality of life. The introduction of the Village Fund in 2015 accelerated the formation of BUMDes across Indonesia. By 2021, approximately 51,134 BUMDes had been established [7] across 74,957 villages. However, only about 30,000 were actively contributing to village income, underscoring the disparity in capacity and operational success. Many attribute this gap to differences in managerial competence, which often result in inefficiencies and limited business development. Legal and administrative issues persist as well. As of 2022, only around 10% of BUMDes had obtained full legal entity status, limiting their access to financing and formal partnerships[8].

I chose East Java province as my sample for three primary reasons. First, it possesses the most comprehensive data on BUMDes establishment, with records dating back to 2001. Second, East Java is recognized as a pioneering province in the creation and utilization of BUMDes. Third, despite being just one of Indonesia's 38 provinces, East Java comprises 8,494 villages out of Indonesia's 83,763, representing approximately 10.1% of the national total. Additionally, the province's population stands at around 41.5 million, making up about 14.5% of Indonesia's estimated 285 million people in 2025[9]. This substantial representation in both village count and population underscores East Java's significance in assessing the impact of BUMDes on rural economic development.

From **Figure 2**, we can see that by 2013 that is just before the announcement of the Village Fund policy, the share of BUMDes in East Java was very low and relatively scattered across the region. By 2015, when the policy was rolled out, there was a noticeable increase in the number of BUMDes created, more evenly distributed across the province. By the latest data in 2025, almost all regions in the province have established BUMDes.

The relative success of BUMDes in East Java is attributed to a strong institutional support from local governments and universities, which offer training and technical assistance; a dense rural population with accessible infrastructure that facilitates market access; and proactive village leadership in leveraging Village Fund allocations to seed and expand local enterprises. However, East Java is not immune to the national challenges faced by BUMDes, including uneven performance across districts, limited managerial

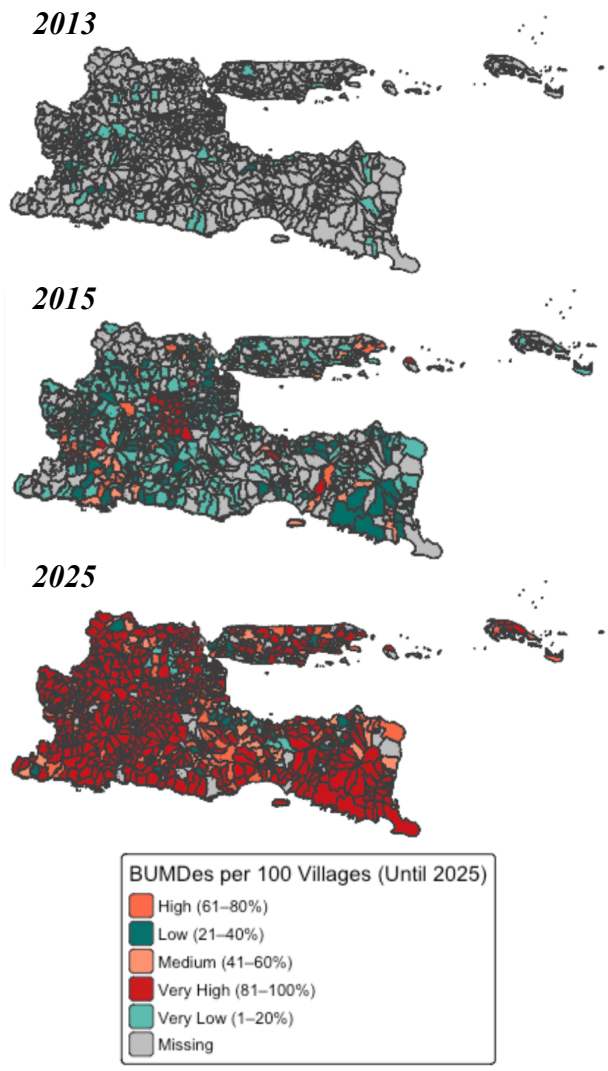
[7] Sofi, I. (2022). Development of Village-Owned Enterprises (BUMDes) in Indonesia. ***Eurasia: Economics & Business***, 10(64), 30–35. https://econeurasia.com/issue-2022-10/article_04.pdf

[8] ibid

[9] Badan Pusat Statistik. (2024). ***Population of Indonesia by Province: Mid-2024 Estimates***. Jakarta: BPS-Statistics Indonesia.

professionalism, and a low rate of legal formalization. In recent years, provincial authorities have intensified efforts to digitize BUMDes operations and integrate them into regional economic and tourism supply chains.

Figure 2. Distribution of Bumdes in East Java in 2013, 2015 and 2025
Source: Data Center of East Java Province



Data

The primary variable of interest in this study is village income, defined as revenue generated locally by the village independently from transfers received from higher government levels. This revenue typically originates from natural resources, local economic potential, and other economic activities managed directly by village governments. Village income thus serves as the key indicator for evaluating economic performance and outcomes at the village level. I use the natural logarithm of village income to normalize the distribution, reduce skewness, and allow for interpretation of the coefficients as approximate percentage changes. Moreover, focusing on percentage change highlights progress more meaningfully than absolute levels, particularly in contexts where baseline income varies widely. For low-income villages, even small absolute gains can represent substantial improvements in economic performance, which would be understated if changes were measured in levels rather than proportionally.

The dataset utilized for this research is sourced from the Village Potential Survey (Podes), a nationally representative survey conducted every three years by Statistics Indonesia (Badan Pusat Statistik – BPS), which ensures standardized and comprehensive data collection across Indonesian villages. The unit of observation in this dataset is the individual village eligible to receive funding under the Indonesian government's Village Fund program, encompassing the complete population of villages targeted by this policy framework. However, the available dataset includes only a single post-treatment observation period, which restricts my ability to apply a staggered

difference-in-differences research design. The potential for this approach could be revisited in future studies once additional post-treatment survey waves become available, thus facilitating a village-by-wave panel analysis.

Data on villages receiving Village Funds are derived from annual reports by the Directorate General of Fiscal Balance, subsequently classified into treatment and control groups, with certain cases excluded based on criteria detailed later in the paper. To robustly analyze variations in village income generation, the study includes additional village-level characteristics potentially correlated with economic performance, such as population size proxied by the number of households, presence of permanent and semi-permanent traditional markets, and availability of educational facilities ranging from primary schools to high schools. Furthermore, the analysis explores heterogeneity in treatment effects based on poverty levels and the educational attainment of village heads. The data analyzed span four survey waves: 2008, 2011, and 2014 serving as pre-treatment observations, and 2019 as the post-treatment observation. All of these data gathered from the same Podes survey.

Descriptive Statistics

Table 1 provides descriptive statistics comparing key characteristics between the villages with and without pre-existing BUMDes before and after the policy intervention. For brevity, I refer to villages with pre-existing BUMDes as 'BUMDes villages' and those without as 'non-BUMDes villages.

Prior to the intervention, the BUMDes villages had a significantly higher average log village income (4.54) compared to non-BUMDes villages (4.10), with a difference of 0.44 that is statistically significant at the 1% level. This baseline imbalance indicates that BUMDes villages were somewhat economically better off even before the policy implementation, suggesting the importance of controlling for baseline income differences in subsequent analyses, which I will focus on this section.

Other pre-treatment village characteristics, including the number of schools, number of families, percentage of villages with semi-permanent and permanent markets, and poverty intensity, show relatively small and statistically insignificant differences between treatment and control groups.

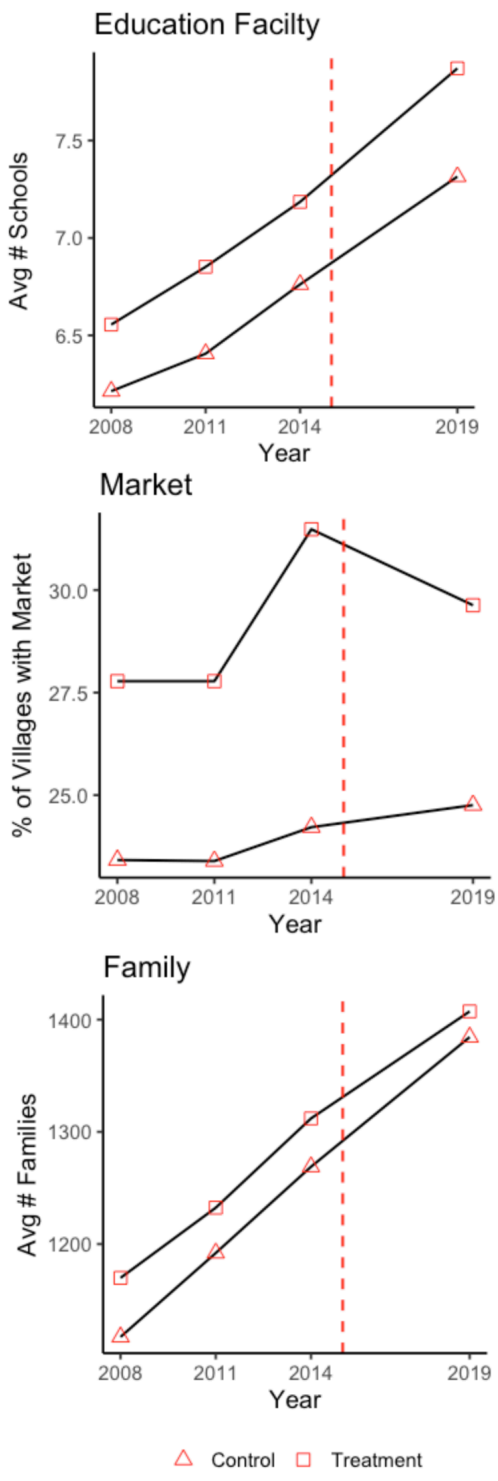
Table 1. Mean (Standard Error) and Differences by Period and Treatment Status

Variable	Pre Treatment	Pre Control	Diff (Pre)	Post Treatment	Post Control	Diff (Post)
Log Village Income	4.54 (0.08)	4.10 (0.02)	0.44***	5.15 (0.19)	4.23 (0.04)	0.91***
# of Schools	6.86 (0.46)	6.46 (0.06)	0.4	7.87 (0.88)	7.31 (0.11)	0.56
# of Families	1238.06 (59.94)	1192.78 (10.56)	45.28	1407.33 (116.12)	1384.41 (20.91)	22.92
% of Village with Market	0.29 (0.04)	0.24 (0.01)	0.05	0.30 (0.06)	0.25 (0.01)	0.05
Poverty Intensity	0.06 (0.00)	0.07 (0.00)	0	0.09 (0.01)	0.10 (0.01)	-0.01

Over the time of this observation, the BUMDes villages had slightly more schools (6.86 vs. 6.46), a higher average number of families (1238 vs. 1193), and a greater percentage of villages with markets (29% vs. 24%), though none of these differences were statistically significant. Poverty intensity was virtually identical across groups (0.06 vs. 0.07). After the intervention, descriptive statistics indicate that the difference in log village income between treatment and non-BUMDes villages widened substantially to 0.91, maintaining statistical significance at the 1% level. This suggests a potential positive effect of the intervention on village income. Other characteristics remained stable, with no statistically significant differences emerging post-treatment. The number of schools, families, the percentage of villages with markets, and poverty intensity exhibited minimal and statistically insignificant variations.

Parallel trends analysis for control variables (see **Figure 3**) further supports the validity of the DiD approach. Trends in the average number of schools, percentage of villages with markets, and average number of families in treatment and non-BUMDes villages closely tracked each other in all three years of observation before the policy intervention in 2015, indicating that both groups exhibited parallel developments, with slight noise in the share of villages with markets right before 2015. This parallel pre-intervention trend strengthens confidence that post-intervention differences, particularly in the main outcome (log village income), are attributable primarily to the policy intervention.

Figure 3. Trend in Control Variables



Another important consideration in this analysis is whether characteristics unrelated to village income but potentially influenced by fiscal transfers, such as school infrastructure changed due to the Village Fund policy. At baseline, the number of schools was similar between villages with and without pre-existing BUMDes, indicating comparable starting conditions. Since school construction is a plausible use of Village Funds, tracking changes in school infrastructure offers a meaningful placebo test.

The observed stability in school numbers across both groups before and after the policy suggests that schools were not meaningfully affected. This supports the use of school infrastructure as a clean placebo variable, helping to test whether the observed income effects are specific to institutional capacity (via BUMDes) rather than broader infrastructural improvements.

These descriptive results reinforce the necessity of controlling for baseline economic differences in subsequent DiD analyses to isolate the policy effect accurately. Moreover, the absence of significant changes in non-income variables strengthens confidence that observed income effects can be attributed more directly to the policy intervention.

Empirical Strategy

I exploit the fact that not all villages had established a Village-Owned Enterprise (BUMDes) prior to the announcement of the policy in 2014. The treatment group is defined as villages that already had a BUMDes in place before 2014 and subsequently received Village Funds, while the control group includes villages that did not have a BUMDes before 2014 but also received Village Funds starting in 2015. In other words, all villages were exposed to the same Village Funds policy intervention, but only treated villages had a pre-existing institutional mechanism (BUMDes) to potentially leverage those funds.

Table 2. Treatment Design Matrix

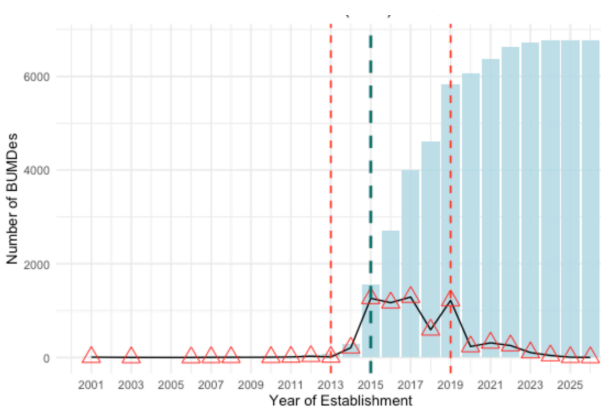
	Pre	Post
VF Treated	0	1
VF Control	0	1
BUMDes Treated	1	1
BUMDes Control	0	0

East Java provincial data indicate that only a fraction of villages had established BUMDes at the policy's outset (77 in 2013, growing to 6780 by 2020), creating substantial initial variation suitable for comparative analysis. Villages that formed BUMDes between 2014 and 2018 are excluded from the main sample for two reasons: first, the dataset includes only one post-treatment period, limiting the feasibility of estimating differential timing effects; and second, late-forming BUMDes are likely influenced by the policy itself, introducing endogeneity concerns. The observed spike in BUMDes creation during 2014–2015 further suggests potential anticipation effects.

To support internal validity, I argue that the decision to form BUMDes before 2014 is plausibly exogenous to both the policy and future outcomes. Treatment status is assigned based on the existence of BUMDes

as of 2013, thus predating the formalization of the Village Fund program. The key identifying assumption is that BUMDes formation prior to 2014 was not systematically driven by foreknowledge of the policy. This is credible for several reasons. Before the Village Fund, there was no large-scale, institutionalized fiscal transfer mechanism at the village level. Hence, it is unlikely that villages strategically established BUMDes in anticipation of such funding. Prior to this, BUMDes had minimal access to state fiscal resources, eliminating strong incentives to form them for rent-seeking purposes. This reduces the likelihood of reverse causality.

Figure 4. Annual Number of BUMDes Formed (Flow) and Cumulative Stock



Evidence from regional studies suggests that early BUMDes were often formed due to local leadership initiatives, NGO involvement, or donor-funded pilot programs (Triwibowo and Aziz, 2020; Rahmawati and Hidayah, 2021; Vel, Fitri, and Cipto, 2020). These idiosyncratic factors are orthogonal to future economic outcomes induced by Village Fund transfers. Taken together, this supports the use of pre-2014 BUMDes presence as a valid treatment variable within a difference-in-differences framework. Since all villages received Village Funds beginning in 2015, but only a subset had institutional capacity in place beforehand, the empirical strategy isolates the interaction between institutional readiness and fiscal transfers, rather than capturing the effect of the transfer itself.

Restricting the sample leads to the identification of a Local Average Treatment Effect (LATE), specifically, the effect for villages that had established BUMDes before 2014, relative to those that either never adopted BUMDes or did so much later. As a result, the findings should be interpreted with caution when generalizing late-adopting villages, as the estimated effects may not extend to those with different institutional trajectories.
Based on this, I have the following reggresion equation:

$$ln(\text{Village Income}_{it}) = \alpha + \beta(\text{Post}_t \times \text{Treat}_i) + X'_{it}\gamma + \vartheta_i + \lambda_t + u_{it}$$

The variable Treat_i is a binary indicator equal to 1 for villages that established a Village-Owned Enterprise (BUMDes) before 2014, and 0 otherwise. The variable Post_t indicates the post-policy period, equal to 1 for years 2015 and thereafter, captured in this study by the year 2019. The parameter of interest, β, captures the DiD effect and reflects the differential change in log village income after the introduction of the Village Fund policy for villages with

pre-existing BUMDes compared to those without.

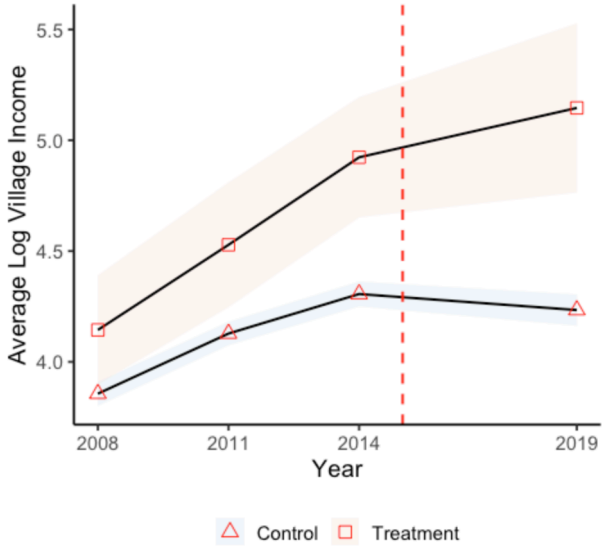
The specification includes village fixed effects (ϑ_i) to control for all unobserved, time-invariant characteristics at the village level, such as geography, initial revenue capacity, or historical institutional quality. Year fixed effects (λ_t) are incorporated to control for shocks or macroeconomic events common to all villages in a given year. By design, the main effects of the treatment and the post-period indicators are subsumed by these fixed effects, ensuring the coefficient β captures purely within-village variations in response to the policy.

Additionally, three village-level, time-varying control variables $X'_{it}\gamma$. They are number of educational facilities, share of villages with market presence, and family count to measure population included to control for evolving village characteristics that might independently affect revenue outcomes and could correlate with treatment.

Standard errors are clustered at the district (kabupaten) level to address potential correlation among villages within the same district. Such clustering is crucial given that villages in close geographic proximity may be influenced by similar district-level policies or economic shocks, resulting in correlated error terms within districts. This approach yields robust inference and guards against underestimated standard errors due to within-district spatial correlation.

Parallel Trend Analysis

Figure 4. Average Log Village Income Over Time by Group (with CI)

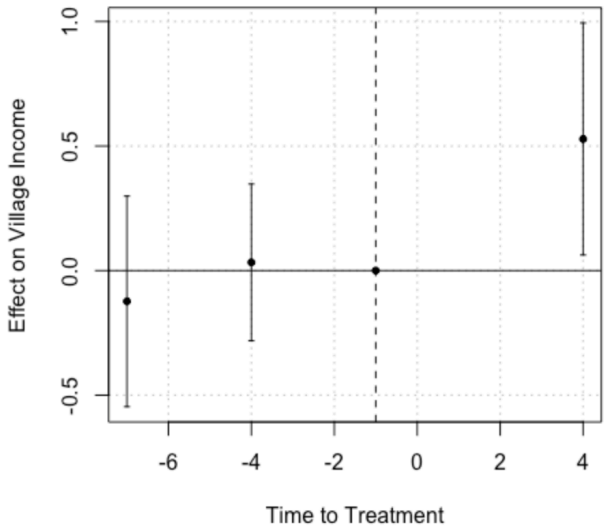


To validate the core identifying assumption of the difference-in-differences (DiD) strategy that in the absence of treatment, villages with and without pre-2014 BUMDes would have followed similar trends in development outcomes, I employ two complementary approaches. I observed visual trend comparison and an event study specification using log of village income as the primary outcome.

From **Figure 4**, a visual inspection of average log village income trends across years (2008–2019) reveals that treated and non-BUMDes villages exhibit broadly parallel trajectories prior to the implementation of the Village Fund policy in 2015. Both groups demonstrate upward movement in log village income, with no evidence of a sharp divergence before the policy year. This graphical evidence suggests that pre-treatment differences were minimal and trends in outcomes

evolved similarly across groups before exposure to the policy intervention.

Figure 5. Event Study: Parallel Trends Check for Log Village Income



To complement this descriptive evidence, I conduct a formal event study analysis. The event study regression **Figure 5** estimates the differential log village income trajectory between treatment and non-BUMDes villages in the years before and after policy implementation, with year -1 (2014) as the reference. The plot shows that the coefficients on the pre-treatment years (years -6 and -4) are close to zero and not statistically significant, with confidence intervals overlapping the zero line. This reinforces the conclusion that there were no significant differences in outcome trends before treatment, satisfying the parallel trends assumption.

Importantly, the event study also detects a clear divergence after treatment: the post-treatment coefficient (year +4) is positive and statistically significant, suggesting that villages with pre-existing BUMDes experienced significantly higher village income growth relative to non-BUMDes villages in the post-policy period. This dynamic response supports the hypothesis that the presence of BUMDes acted as an institutional amplifier for the Village Fund transfers.

Taken together, the parallel pre-treatment trajectories observed in both the visual trends and event study regression provide strong empirical support for the internal validity of the DiD approach. The findings lend credibility to a causal interpretation of the estimated impact of BUMDes on log village income outcomes following the introduction of Village Funds.

Findings

Table 3 presents the DiD regression estimates of the impact of BUMDes on log village income. I report several specifications to check robustness: (1) with no controls, (2) adding population (family) control, (3) adding educational facilities, and (4) adding market presence. All models include village and year fixed effects and are clustered by district.

Across all specifications, the coefficient on the Treat × Post interaction is remarkably stable and statistically significant at the 5% level. In the fullest model (column 4), the estimate of β is around 0.530 (with a standard error such that $p < 0.05$). This indicates that villages with an early BUMDes saw about a 53% higher increase in log village income after the Village Fund policy relative to villages without a BUMDes. In other words, the presence of a BUMDes amplified the growth of village

revenue by nearly one-half in proportional terms. This is a sizable effect, suggesting that the institutional capacity provided by a BUMDes enabled villages to better convert fiscal transfers into income-generating activities, consistent with previous literature (Sambodo et al., 2024; Ultari & Khoirunurrofik, 2024).

Table 3. DiD Regression Results

	(1)	(2)	(3)	(4)
Treatment × post	0.551**	0.537**	0.532**	0.530**
	(0.222)	(0.220)	(0.220)	(0.219)
Family		0.000***	0.000**	0.000**
		(0.000)	(0.000)	(0.000)
EduFacility			0.008	0.005
			(0.011)	(0.012)
MarketPres				0.133
				(0.119)
Num.Obs.	7465	7465	7465	7465
R2	0.511	0.516	0.516	0.516
R2 Adj.	0.381	0.387	0.387	0.388
Village Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Clustered SE (KAB Level)	Yes	Yes	Yes	Yes

* p <0.1, ** p <0.05, *** p <0.001

It is worth noting that a 53% increase in log village income is a percentage increase; given that baseline village income levels were quite low in many villages, this effect, while large in percentage terms, corresponds to a modest absolute revenue gain (on the order of a few million rupiah on average). Nonetheless, for village governments with tiny budgets, even a few extra million rupiah can be important for local service provision.

The control variables behave as expected, and their inclusion does not materially change the treatment effect. The coefficient on the Family (proxy for population) variable is positive and highly significant (though very small in magnitude, on the order of 0.0003), indicating that larger villages have slightly higher village income (each additional family adds a negligible fraction to log village income). Educational facilities show a negative but insignificant coefficient, perhaps suggesting that having more schools (often an indicator of a more developed village) does not directly translate to higher village income. Market presence enters with a positive sign, implying that villages with a market tend to have higher village income, but this effect is not statistically significant in my sample. Importantly, adding these controls does not reduce the magnitude or significance of β, which remains at around

0.53. This robustness gives confidence that my main result is not driven by omitted observable differences.

In summary, the regression evidence supports the hypothesis that having a BUMDes is associated with different trajectories in village revenue generation between the treatment and control groups after the Village Fund policy. The consistency across models and the alignment with the event-study findings suggest that BUMDes acted as an enabling mechanism that helped villages better leverage the inflow of central funds, although the results should not be interpreted as definitive causal effects.

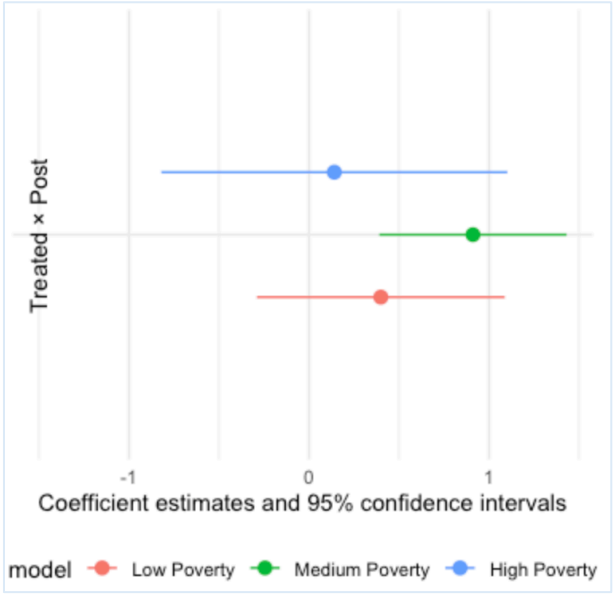
These findings connect back to the broader literature: they provide micro-level confirmation that institutional capacity matters. While Hilmawan et al. (2023) found a weak overall impact of BUMDes on a composite development index, my results show a strong association with a specific outcome (local revenue), which is logically the direct channel through which a BUMDes would affect development. This suggests that BUMDes do fulfill their intended role of raising village income, even if those gains might take time or further support to translate into broader development outcomes.

Heterogenous Effects

I explored heterogeneity in the treatment effect to understand in what contexts BUMDes are most effective. Specifically, I examine whether the impact varies by (a) baseline poverty level, (b) village head's education, and (c) village population size. These factors were chosen based on theoretical considerations and anecdotal evidence suggesting they influence institutional performance.

Impact by Poverty Level

Figure 6. Heterogeneous Treatment Effects by 2014 Poverty Intensity Level



I stratify villages into tertiles of the 2014 poverty intensity ratio (low, medium, high poverty) and re-run the DiD regression separately for each group. I choose the year 2014 to assess these villages right before they receive the Village Fund.

From **Figure 6,** all three poverty groups exhibit a positive treatment effect (BUMDes coefficient) on log village income, but the strength and precision differ notably. The impact is strongest and most precisely estimated for the middle tertile. These villages show a coefficient similar to the full sample, and it is statistically

significant. The confidence interval is relatively narrow, indicating a robust effect. I interpret that in villages with moderate poverty (not the poorest, but not wealthy), a BUMDes was particularly effective, possibly because there is enough economic base to leverage and sufficient need or motivation to utilize the BUMDes, consistent with previous literature review (Chaudhuri et al., 2019)

The poorest villages also have a positive coefficient, but it is slightly smaller and statistically significant at a lower confidence level (10%). The confidence interval is wider. This suggests that in very poor villages, while BUMDes did help increase village income, the effect was more variable. Some high-poverty villages with BUMDes may have still struggled due to structural challenges (e.g., low human capital, remote location), yielding heterogeneity in outcomes. This is echoing earlier findings in intervention on extreme poverty in India (Bardhan & Mookherjee, 2006). Still, the point estimate is meaningful, implying even poor villages can see gains with the right institution.

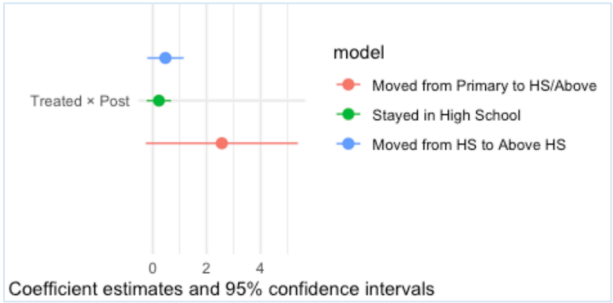
Interestingly, the treatment effect is weakest for low-poverty (relatively wealthy) villages. The point estimate is positive but much smaller, and it is statistically insignificant, with a wide confidence interval that includes zero. This imprecision may indicate that richer villages did not benefit as much from BUMDes, perhaps because they already had other sources of revenue or less "low-hanging fruit" to exploit. In affluent villages, Village Funds might represent a smaller part of the budget, and a BUMDes might have marginal utility, as also pointed out by previous research finding (Stokan & Overton, 2024)

These patterns suggest a U-shaped relationship: BUMDes had the largest impact in middle-income contexts, with somewhat lower impacts at both extremes (though still positive in high-poverty areas). One interpretation is that absorptive capacity and need are both important. Very poor villages have high needs but may lack the capacity to effectively run enterprises (yet some still succeed, hence an average positive effect with more variance). Richer villages have capacity but less urgent need or fewer untapped opportunities. Those in the middle have a balance of need and capacity.

Impact by Village Head's Education

I next consider the role of leadership quality, proxied by the education level of the village head.

Figure 7. Heterogeneous Treatment Effects by 2014 Village Head's Education



I categorize villages based on the change in their village head's education between 2014 and 2019: one group where the head's

education remained low (primary or below in both years), another where the head improved from primary in 2014 to at least high school by 2019, a third where the head consistently had high school, and a fourth where the head went from high school to above high school. This captures whether an improvement in leadership human capital correlates with better use of BUMDes.

Leadership capacity, proxied by village head education levels, significantly influenced the observed trajectories. Villages led by heads who moved from high school to above high school between 2014 and 2019 experienced the strongest differences in trajectories. Conversely, villages consistently led by heads who stayed at high school level also realized positive and statistically significant differences, though slightly smaller. Villages where the head moved from primary to high school or above showed positive point estimates but with very wide confidence intervals crossing zero, indicating that the differences were statistically imprecise and not significantly different from zero. This suggests that upgrading leadership education from already relatively high levels (high school to above high school) was associated with stronger revenue generation outcomes, while improvements from primary levels were more variable and less reliably linked to enhanced village income trajectories.

These findings underscore the importance of human capital and governance in making village enterprises work. They resonate with observations that many BUMDes fail to thrive due to limited managerial skills. When a village head is better educated or gains training (as some did as part of the Village Law implementation), the BUMDes is more likely to succeed. This finding strengthen previous study that associates higher level of education of village heads with village development index (Mahi, 2023). For policymakers, this implies that complementing fiscal transfers with investments in local leadership and managerial capacity can pay dividends.

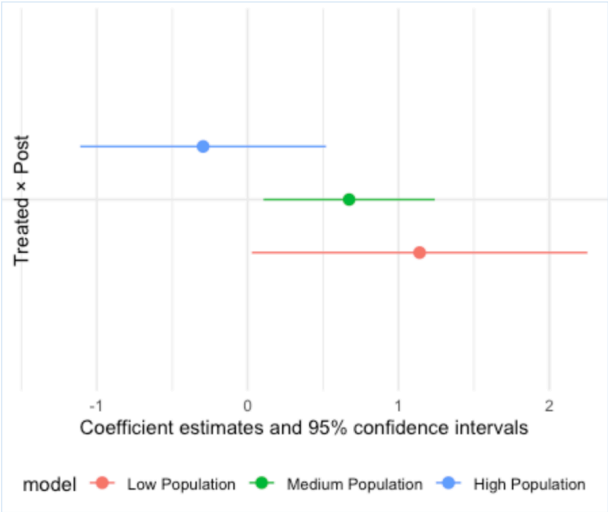
Impact by Village Size

Finally, I examine whether village size (population) influences the effectiveness of BUMDes. I split the sample into tertiles by the average number of families (households) in 2014 and 2019: small villages, medium-sized, and large villages. Intuitively, larger villages might have more economic activity and larger markets to support a BUMDes venture, but they might also face more competition or complexity. My results show that the BUMDes treatment effect is positive in low- and medium-population villages but inconclusive in large-population villages, with differences in precision.

Large villages (top tertile) exhibit an inconclusive impact, with a wide confidence interval crossing zero. Medium-sized villages likewise show a positive point estimate, but the estimate has a wider confidence interval and is slightly less precise. Small villages (bottom tertile) also experience a positive BUMDes impact, and the estimate is relatively more precise compared to medium-sized villages. This indicates greater uncertainty and

variability among medium-sized villages, with some may have benefited greatly from a BUMDes, while others perhaps not at all. One reason could be that in moderately sized communities, the scope of business opportunities varies widely, or a single idiosyncratic factor (like a particularly dynamic or poor-performing BUMDes) can swing the results. In contrast, small villages likely have more consistent but modest opportunities to generate returns from a BUMDes, and the law of large numbers applies less.

Figure 8. Heterogeneous Treatment Effects by Population Size (2014 & 2019)



Population size modulates the consistency (and in some cases the direction) of the BUMDes effect. Low- and medium-population villages show that BUMDes helped increase village income, while large villages show inconclusive results with confidence intervals crossing zero, which is important to consider for generalizability. This is consistent with the previous finding examining the effect of same policy by population Muhlisin et al., 2024). Supporting BUMDes in larger villages might require extra attention to ensure viability.

Conclusion

This study provides robust evidence that institutional capacity, in the form of having BUMDes, can significantly enhance the effectiveness of large-scale fiscal decentralization policies. This DiD analysis shows that villages that established BUMDes before the announcement of the policy experienced a 53% higher increase in log village income compared to those that did not. This result remains statistically significant and stable across all model specifications that I tested, reinforcing the notion that institutional capacity meaningfully shapes development trajectories.

These findings have two key policy implications: institutional capacity and readiness, as well as village characteristics, matter. The Village Fund is a powerful decentralization policy in enabling village governments to develop their local economies, but without a certain level of capability and appropriate tools to manage the funds, its impact remains limited. Interventions that incentivize villages to build capacity such as providing training to BUMDes officers and standardizing practices are necessary to ensure that the large number of BUMDes created after the implementation of the policy can match the capabilities of those established before the policy. The latter emerged from the villages' needs and capacity to manage

enterprises, rather than being created merely to comply with policy incentives.

Second, different baseline characteristics of BUMDes villages, those argued to possess institutional capacity, also yielded varied results. Medium-poverty villages performed better in village income outcomes, likely because they possessed both the need and the capacity to implement and sustain initiatives encouraged by BUMDes. Addressing persistent poverty, which remains more severe in some villages, is crucial prior to disbursing fiscal transfers. More tailored and targeted fund allocation for low-poverty villages is also needed to ensure the policy's impact. Similarly, villages with smaller populations showed more precise results, likely because the per capita intensity of the transfers is higher compared to larger villages that receive similar amounts, resulting in significantly lower per capita funding. Policymakers need to recognize that these village characteristics matter. Different villages should be treated differently, or at least policy should be adjusted according to their needs and structural context. My results also highlight that investments in leadership and governance capacity such as training for village heads can further amplify the effectiveness of fiscal programs, especially in rural contexts.

However, several limitations remain. The analysis relies on only one post-treatment period, which restricts my ability to capture longer-term dynamics or implement staggered treatment designs. The results, while compelling, represent a Local Average Treatment Effect (LATE). They speak primarily to villages that had BUMDes before 2014 and may not generalize to late adopters or those yet to establish such institutions. Measurement error and omitted variable bias, particularly related to leadership quality in the treatment group and village-level social networks, may also threaten internal validity. I also acknowledge that even within the treatment group, the capacity and capability of BUMDes likely differ, resulting in varied treatment intensities across villages. I do not have data on these BUMDes characteristics, which limits my ability to explore these variations.

Future research should extend the time frame to include more post-treatment waves, enabling dynamic analysis of the long-term sustainability of BUMDes. Incorporating detailed BUMDes-level data, such as the education level of BUMDes leaders, annual financial performance, business type and scale, and other enabling factors would be highly valuable in understanding to what extent BUMDes directly influence village income.

Ultimately, these results underscore the importance of aligning financial decentralization with institutional readiness, a lesson that extends beyond East Java, or even Indonesia to any country confronting the challenge of building state capacity from the ground up to narrow the rural-urban development gap and, eventually, alleviate poverty.

References

Adam, L., & Hidayat, S. (2023). ***Village fund and Indonesia rural development: Rethinking institutional capacity and governance***. Journal of Infrastructure, Policy and Development, 7(1), 100123.

Adam, L., Sambodo, M. T., Purwanto, P., Purnagunawan, R. M., Sarana, J., Ermawati, T., et al. (2024). ***Village fund and Indonesia rural development: Rethinking institutional capacity and governance***. Journal of Infrastructure, Policy and Development, 8(1), 9875.

Agrawal, A., & Ribot, J. (1999). ***Accountability in decentralization: A framework with South Asian and West African cases***. The Journal of Developing Areas, 33(4), 473–502.

Bardhan, P., & Mookherjee, D. (2005). ***Decentralizing antipoverty program delivery in developing countries***. Journal of Public Economics, 89(4), 675–704. https://doi.org/10.1016/j.jpubeco.2003.01.001

Bardhan, P., & Mookherjee, D. (2006). ***Decentralization and accountability in infrastructure delivery in developing countries***. The Economic Journal, 116(508), 101–127. https://doi.org/10.1111/j.1468-0297.2006.01075.x

Bardhan, P., & Mookherjee, D. (2006). ***Pro-poor targeting and accountability of local governments in West Bengal***. Journal of Development Economics, 79(2), 303–327. https://doi.org/10.1016/j.jdeveco.2005.03.002

Chaudhuri, E., Kurniawati, S., & Sumarto, S. (2019). ***Decentralisation and poverty reduction: The role of local economies and institutional capacity in Indonesia*** (Working Paper No. 45/2019). Jakarta: TNP2K. Retrieved from https://www.tnp2k.go.id

Faguet, J. P. (2014). ***Decentralization and governance***. World Development, 53, 2–13. https://doi.org/10.1016/j.worlddev.2013.01.002

Faguet, J. P., & Pöschl, C. (Eds.). (2015). ***Is decentralization good for development? Perspectives from academics and policy makers***. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780198737504.001.0001

Government of Indonesia. (2014). ***Law No. 6 of 2014 on Villages***. https://faolex.fao.org/docs/pdf/ins161827.pdf

Grindle, M. S. (2007). ***Going local: Decentralization, democratization, and the promise of good governance***. Princeton University Press.

Hilmawan, R., Prakasa, B., & McCarthy, J. F. (2023). ***Village-owned enterprises and rural transformation: A mixed-methods study from Indonesia***. World Development Perspectives, 32, 100503. https://doi.org/10.1016/j.wdp.2023.100503

Mahi, B. R. (2023). ***Village development: Effect of Village Fund and village head education***. Economics Development Analysis Journal, 12(2), 141–156. https://doi.org/10.15294/edaj.v12i2.69091

Muhlisin, Subroto, W. T., & Sakti, N. C. (2024). ***Village development and welfare enhancement: The influence of area size, population, and Village Fund transfer (Evidence from Bangkalan, Indonesia)***. International Journal of Research and Scientific Innovation, 11(10), 433–443. https://doi.org/10.5120/ijrsi202411039

Permatasari, P., Budiarso, A., Dartanto, T., Samosir, A. P., Saputro, B., Ekayana, D., Agustono, E. D., Alim, T. E., Hartono, L., Wahyuputri, F. W., & Wardhana, I. W. (2024). ***Village fund management and reporting systems: Are they accountable?*** Transforming Government: People, Process and Policy, 18(4), 512–528. https://doi.org/10.1108/TG-07-2023-0098

Rahmawati, Fitri, and Sulistyaningtyas Hidayah. 2021. "Evaluasi Pengelolaan BUMDes sebagai Badan Usaha Milik Desa di Kabupaten Bojonegoro." ***Jurnal Ilmiah Ilmu Administrasi Publik*** 11(2): 199–210;

Smoke, P. (2003). ***Decentralisation in Africa: Goals, dimensions, myths and challenges***. Public Administration and Development, 23(1), 7–16. https://doi.org/10.1002/pad.255

Stokan, E., Deslatte, A., & Overton, M. (2024, May 7). ***Neediest areas are being shortchanged on government funds – even with programs designed to benefit poor communities***. The Conversation. https://theconversation.com/neediest-areas-are-being-shortchanged-on-government-funds-227074

Triwibowo, Dandy Koswaraputra, and Fakhrudin Aziz. 2020. ***Studi Pembelajaran Inisiatif Membangun Desa: Pembentukan BUMDes di Kabupaten Batanghari***. Jakarta: The Asia Foundation

Ultari, T., & Khoirunurrofik. (2024). ***The role of Village-Owned Enterprises (BUMDes) in village development: Empirical evidence from villages in Indonesia***. Jurnal Perencanaan Pembangunan: The Indonesian Journal of Development Planning, 8(2), 256–280. https://doi.org/10.36574/jpp.v8i2.427

Vel, Jacqueline, Nurul Fitri, and Bambang Cipto. 2020. "The Emergence of Village-Owned Enterprises in Indonesia: Local Governance and the Role of External Actors." ***ASEAS – Austrian Journal of South-East Asian Studies*** 13(1): 49–68.

Wunsch, J. S. (2001). ***Decentralization, local governance and 'recentralization' in Africa***. Public Administration and Development, 21(4), 277–288. https://doi.org/10.1002/pad.185

Yeni, F. (2021). ***The role of social capital on rural enterprises economic performance: A case study in Indonesian villages***. SAGE Open, 11(3), 1–10. https://doi.org/10.1177/21582440211044178